

Counting Real Roots in Polynomial-Time for Systems Supported on Circuits

J. Maurice Rojas

June 9, 2021



What is the *fine-grained* complexity of counting real roots of sparse polynomial systems?



Counting Real Roots in Polynomial-Time

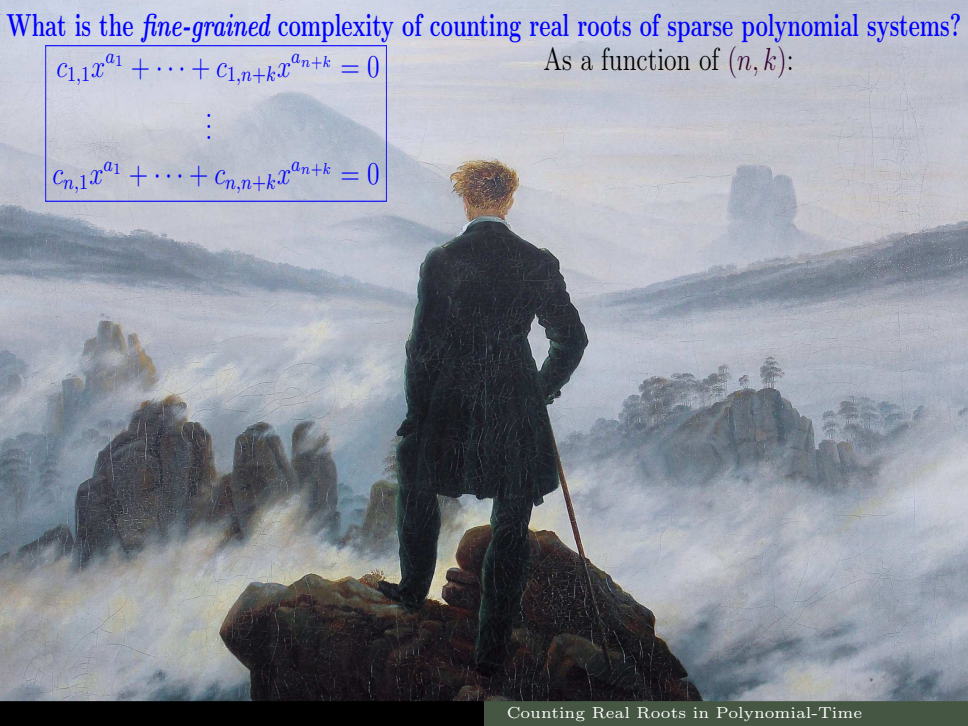
What is the *fine-grained* complexity of counting real roots of sparse polynomial systems?

As a function of (n, k) :

$$c_{1,1}x^{a_1} + \cdots + c_{1,n+k}x^{a_{n+k}} = 0$$

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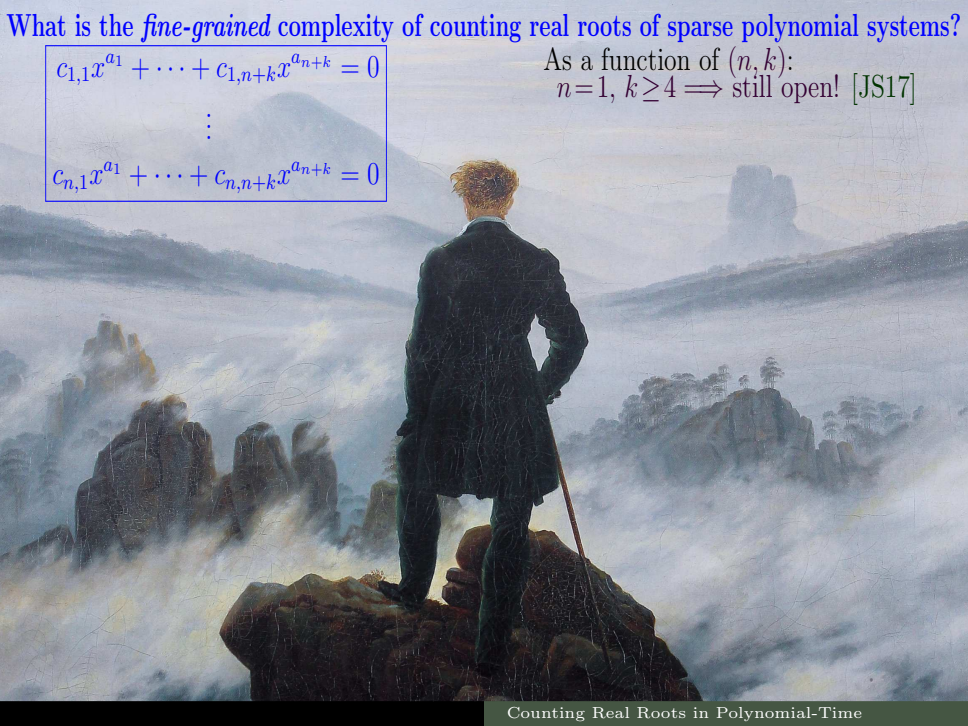
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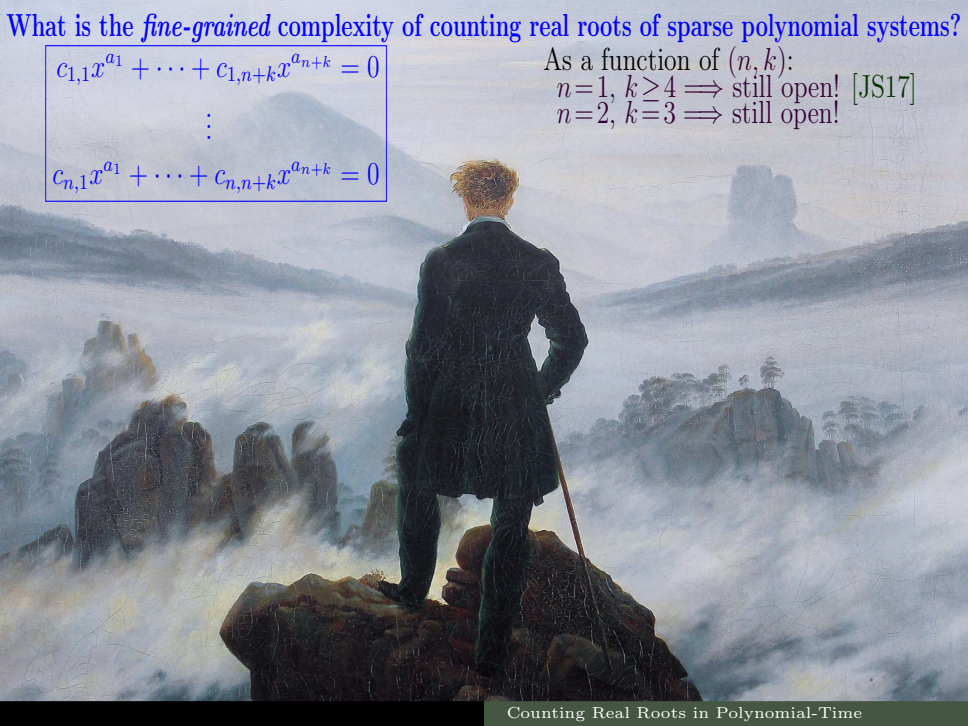
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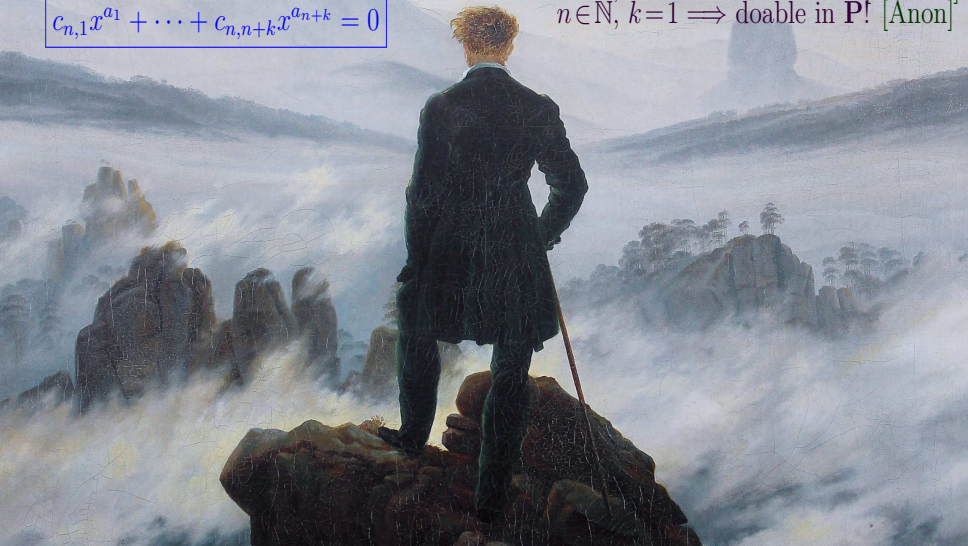
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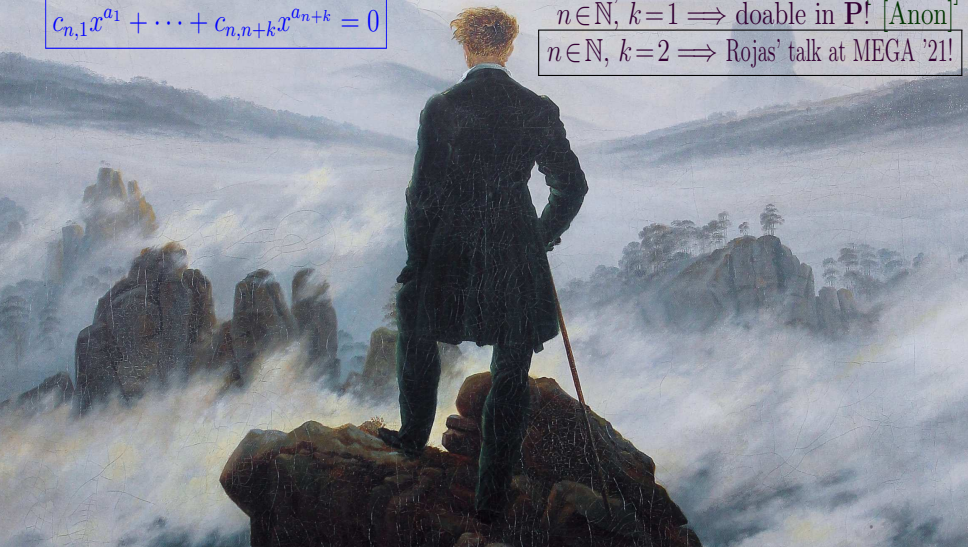
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$n \in \mathbb{N}, k=2 \implies$ Rojas' talk at MEGA '21!



Main Theorem (coarsely)

[Rojas, 2020] *For any fixed n , one can exactly count the roots in $\mathbb{R}_+^n$, $(\mathbb{R}^*)^n$, and $\mathbb{R}^n$ of any generic $(n+2)$ -nomial $n \times n$ polynomial system over $\mathbb{Q}$ in deterministic polynomial-time.*



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Example: *In < 1 second, we can see that the 7-nomial 5×5 system defined by*

$$\begin{aligned} & \left(2x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 37137cx_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{9}{2}x_3^{166}x_4^{68}x_5^{343}, \right. \\ & x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + 2x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 24849cx_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343}, \\ & x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + 2x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 21009cx_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343}, \\ & x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + 2x_1^{25}x_2^{203}x_3^{44}x_4 + x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 20769cx_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343}, \\ & \left. x_1^{36}x_2^{194}x_3^{50}x_4^{82}x_5^{60} + x_1^{76}x_2^{240}x_4^{41}x_5 + x_1^{74}x_2^{179}x_3^{25}x_5^{57} + x_1^{25}x_2^{203}x_3^{44}x_4 + 2x_1^{20}x_2^{167}x_3^{64}x_4^{12}x_5^{68} - 20754cx_1^{58}x_2^{194}x_3^{24}x_4^{36}x_5^{25} - \frac{21}{4}x_3^{166}x_4^{68}x_5^{343} \right). \end{aligned}$$

has exactly 2, 6, 6, 2, 2, or 0 positive roots, respectively when c is $\frac{1}{20731}, \frac{1}{20730}, \frac{1}{14392}, \frac{1}{14391}, \frac{1}{13059}, \frac{1}{13058}$. $\diamond$



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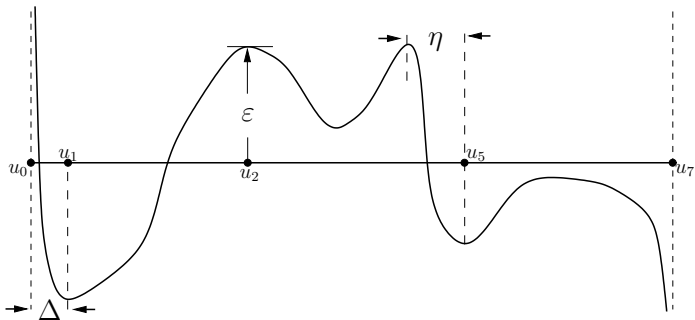
Over 245 million complex roots!

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Key Ideas: Critical Points and Diophantine Approximation

Consider a linear combination of logarithms extracted from your system:

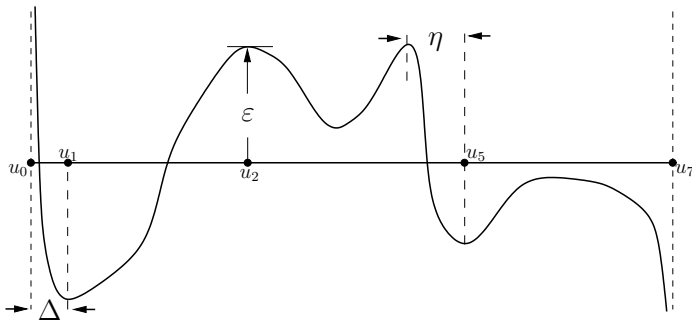


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For details, see Rojas' talk on Wednesday, June 9!

